Functions In Calculus

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1 Introduction

To "compile" this information I am using the book Calculus by David Patrick it can be found here: `https://artofproblemsolving.com/store/book/calculus`.

This Document is designed to review the Function principals covered in the aformentioned book. I will create other "Guides" for other topics covered. They can be found here: `https://github.com/KovasMcCann/Writings`.

2 What is a Function

A function is a matamatical device that associates an input with a unique output.

2.1 Example

$f(x) = 3x - 5$

Input: $x \in \mathbb{R}$
Output: $3x - 5 \in \mathbb{R}$

2.2 Formal Definition

A function f from set A to set B
$f : A \to B$

Where:
$a \in A$ and $f(a) \in B$

3 Properties of Functions

There are three main properties of functions: **Domain**, **Codomain**, and **Range**.

- The **Domain** is the set of all possible input values of a function. It is denoted by $Dom(f)$.

- The **Codomain** is the set of all potential output values defined for the function, whether or not they are actually produced. It is denoted by $Cod(f)$.
- The **Range** is the set of all actual output values that the function produces from the domain. It is denoted by $Rng(f)$.

3.1 Example

Let $f(x) = x^2$, where $x \in \mathbb{R}$.

- $Dom(f) = \mathbb{R}$
- $Cod(f) = \mathbb{R}$
- $Rng(f) = [0, \infty)$

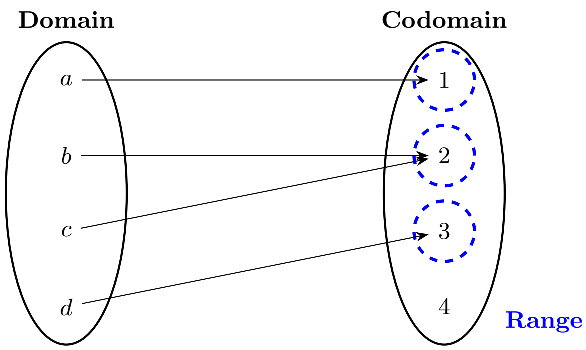


Figure 1: The relationship between input and output in a function. The first set represents the domain (inputs), the second set represents the codomain (all possible outputs), and the blue dashed circles highlight the actual outputs — the range.

4 Operations

All basic arithmatic operations can be applied to functions themselves.

Given two functions $f(x)$ and $g(x)$, we can define the following operations:

- **Addition:**
$$(f + g)(x) = f(x) + g(x)$$
The resulting function represents the pointwise addition of f and g.
- **Subtraction:**
$$(f - g)(x) = f(x) - g(x)$$
This represents the pointwise subtraction of g from f.

- **Multiplication:**
$$(f \cdot g)(x) = f(x) \cdot g(x)$$
The multiplication is also performed pointwise.

- **Division:**
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{provided that } g(x) \neq 0$$
Division is defined only at points where $g(x) \neq 0$.

- **Scalar Multiplication:**
$$(c \cdot f)(x) = c \cdot f(x)$$
where c is a constant. This scales the output of the function by c.

- **Composition:**
$$(f \circ g)(x) = f(g(x))$$
This represents function composition, where the output of g becomes the input to f.

5 Inverse Functions

The inverse of a function is where the X and Y values are swaped. The inverse of a function is denoted by f^{-1}

Example:

$$f(x) = 3x - 5$$

$$f^{-1}(x) = \frac{x+5}{3}$$

5.1 How to Find the Inverse of a Function

To find the inverse of a function, follow these steps:

- Replace each occurrence of $f(x)$ with y
- Swap x and y in the equation
- Solve the resulting equation for y
- Replace y with $f^{-1}(x)$

Example: Find the inverse of $f(x) = 5x - 2$

$$
\begin{aligned}
y &= 5x - 2 \\
x &= 5y - 2 && \text{(Swap } x \text{ and } y\text{)} \\
x + 2 &= 5y && \text{(Add 2 to both sides)} \\
\frac{x+2}{5} &= y && \text{(Divide both sides by 5)} \\
f^{-1}(x) &= \frac{x+2}{5}
\end{aligned}
$$

6 Graphs of Functions

Function's are typically graphed on the **Cartesian plane**.

The Cartesian plane is the set of all ordered pairs of real numbers; i.e:

$$\{(x, y) : x, y \in \mathbb{R}\}$$

Noramally its depicted either like $\mathbb{R} \times \mathbb{R}$ or $\mathbb{R}^2$.

7 Functions Vs. Equations

The defintion of a function and equation are:

A function has at least 2 variables: an output variable and one or more input variables. An equation states that two expressions are equal, and it may involve any number of variables (none, one, or more). A function can often be written as an equation, but not every equation is a function.

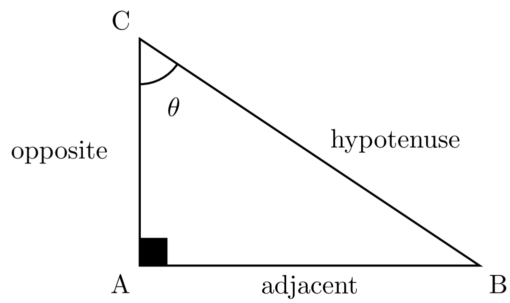
This is becuase functions use the X and Y planes as input and output where $X(Dom())$ is the input and $Y(Cod())$ is the input

Keep this in mind while thinking about Functions.

8 Trigonometic Functions

8.1 Trigonometric Functions of a Right Triangle

These are the typical $sin()$, $cos()$ and $tan()$ functions



Trigonometric Definitions:

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} \quad \cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

8.2 Radians

In calculus measurement of angles is done in radians this is beucase radians are more natural than the arbitrary 360°.

8.2.1 History of Degrees

The history of degrees, while not fully known, is quite interesting. According to Wikipedia:

> The original motivation for choosing the degree as a unit of rotations and angles is unknown. One theory suggests it is related to the fact that 360 is approximately the number of days in a year. Ancient astronomers observed that the sun appears to advance about one degree along the ecliptic path each day. Some ancient calendars, such as the Persian calendar, used 360 days in a year. The use of a 360-day calendar may be connected to the adoption of sexagesimal (base-60) numbers.
>
> Another theory proposes that the Babylonians subdivided the circle using the angle of an equilateral triangle as a basic unit and further divided it into 60 parts, consistent with their sexagesimal numeric system.

8.2.2 What are Radians

Radians are the standard unit for measuring angles in mathematics and physics. Unlike degrees, radians directly relate an angle to the geometry of a circle, making many formulas simpler and more natural.

A radian is defined as the angle formed at the center of a circle when the arc length is equal to the radius:

$$\text{radian} = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

Because both arc length and radius are lengths (e.g., meters), their units cancel out. This makes radians a **dimensionless** unit — it's just a ratio of two lengths.

8.2.3 Where Does π Come In?

π appears naturally in radians because it's part of the formula for the circumference of a circle:

$$\text{Circumference} = 2\pi r$$

So, if you go all the way around a circle, the arc length is $2\pi r$, and using the radian formula:

$$\theta = \frac{s}{r} = \frac{2\pi r}{r} = 2\pi$$

That means a full circle is exactly $\boxed{2\pi \text{ radians}}$, which is about 6.28 radians. For a unit circle (radius 1), the circumference is just 2π, so the full angle around the circle is 2π radians.

8.2.4 Conversions

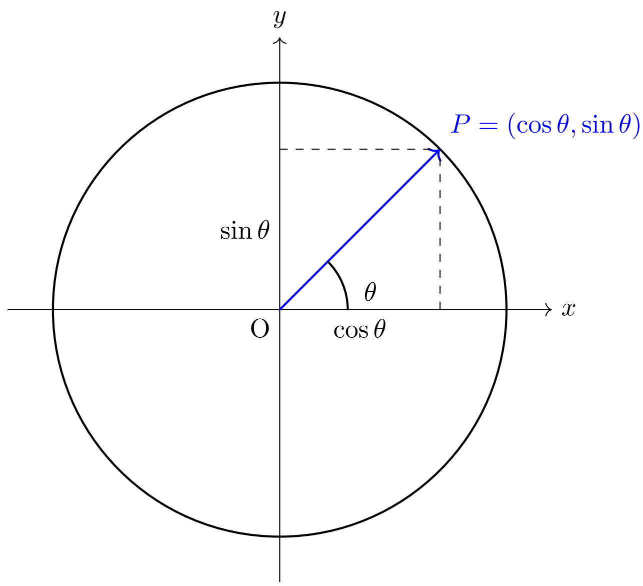
To convert between radians and degrees, use the following formulas:

$$\text{Radians} = \text{Degrees} \times \frac{\pi}{180} \qquad \text{Degrees} = \text{Radians} \times \frac{180}{\pi}$$

8.3 Unit Circle

Now how can we tie in the two ideas of Radians and Trigonometics into one. Well the solution is the **Unit Circle!**

The Unit Circle has a radius of 1 and the circumfrence is denoted by using (X, Y) points.



8.4 Reciprocals

$$\sec(\theta) = \frac{1}{\cos(\theta)} \quad \csc(\theta) = \frac{1}{\sin(\theta)} \quad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

8.5 Inverse Functions

The inverses of the Trigonometric functions swap the input and output. Because of this, they compute the angle of a radius

There or denoted with a $^{-1}$. Example:

$$\sin^{-1}$$

8.6 Trigonometric Identities

Trigonometric identities are equations involving trigonometric functions that are true for all values of the variable where both sides are defined. They are essential for simplifying expressions and solving trigonometric equations.

1. Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1 \quad 1 + \tan^2\theta = \sec^2\theta \quad 1 + \cot^2\theta = \csc^2\theta$$

2. Reciprocal Identities

$$\csc\theta = \tfrac{1}{\sin\theta} \quad \sec\theta = \tfrac{1}{\cos\theta} \quad \cot\theta = \tfrac{1}{\tan\theta}$$

3. Quotient Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

9 Exponentials and Logarithms

9.1 The relationship between Exponentials and Logarithms